



## Bokeh Introduction Transcript

Hello everyone, welcome to Bitsplanet Digital. Today we will continue our learning journey for the course Data Visualisation Storytelling. We will be now looking at a specialised suite called Bokeh, which is a library in the Python ecosystem that helps you create interactive visualisations specifically for web browsers.

Bokeh is a suite that helps you create beautiful data visualisations ranging from simple plots to complex dashboards, which can also feature the live streaming data. You can build very sophisticated visualisations and embed them as standalone HTML objects, or you can also have as a server application, but it can work as simple as a Jupyter notebook. So we will now look at the Bokeh features for creating interactive visualisations.

Let me take a minute here and explain how this fits into the overall visualisation storytelling concept. If you remember, the main objective of a data visualisation course is to make you understand that you have to tell your data story in a way that captivates your audience and empowers them to take the actions that you intend them to do so. One of the main ways to achieve this is to control the narrative in which you are telling your data story.

And the most effective way of creating a narrative that has a balanced view for both authors and the audience. If you recall, the narrative should be such that it is author-driven in the initial stages with features for the audience to interact and make their own conclusions. So in this way, using Bokeh features, you can create interactive data visualisations, which are very striking at the same time, giving you the flexibility for the audience to derive their own conclusions from the data.

So let us now understand the features of Bokeh library in the Python system and then try certain examples. Like in last classes, I will be simultaneously running the Python code so that you can practise them later. So we are going to instal Bokeh.

So always understand this pip instal command, which helps you instal certain packages within the Colab notebook or Jupyter notebook. So now I have installed this in the Colab environment. So I have the necessary features.

So now then I'm going and it runs like any other command that you have in Python. So from already I have loaded Bokeh. So I'm using output notebook, which is a feature that is required for building data visualisations.

So I'm opening and I have done that. Now I've got a notebook where I can work for graphics. So for using Bokeh in Jupyter, you need this output notebook function, which means it creates a corporate workspace for using the Bokeh functions.



Now, let us understand what are components in the Bokeh library or Bokeh system. So Bokeh allows you to use Python built interactive visualisations that run in a web browser. So it includes a JavaScript at the back end.

And this helps you to once you have a JavaScript, there is more of customisation and interactivity can be built in. And then Bokeh has a very good strong documentation and the user guide, which can be looked at directly from their website. So you have a lot of support material from the direct Bokeh website and also a lot of visualisations that are available for you to explore.

So as in last classes, be it the case of Mathplotlib or in Seaborn, we are exploring only certain examples, but you are advised to explore the source documentation and try out certain examples or take certain tutorials here so that you will be able to use the best features depending upon the use case at hand. OK, we will not be able to cover the entirety of these very vast tools, but give an introduction to them so that you will be able to pick up the basics and build from there for whatever applications you are doing in your coursework or in your work environment. Now, let us understand how Bokeh actually works.

There are two ways in which Bokeh features can be used. One is using high level Bokeh plotting interface. Another is low level Bokeh models.

Bokeh's plotting is the primary interface in which Bokeh will produce the graphics. It is a general purpose interface where there are inbuilt libraries like Mathplotlib or Matlab. In this feature, Bokeh plotting, it takes care of many attentive attributes or colouring or styling scheme automatically.

So this makes you to get the data quickly turned into a visualisation, but in a standardised format. But at times you might want to create your own customised visualisations. In that case, you might use Bokeh models, which will allow you to have further control on the customisation of the plots.

OK, so we are focussing on Bokeh plotting interface and objects on the Bokeh models will be used whenever necessary. And you can try to get this from the Bokeh guide if you want any further information or further details.

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